

Hölder and Zygmund Spectra of the Rvachev–Thue–Morse Operator

Finite Fourier reduction, critical Jordan chains,
and optimal decay

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Principal conclusion. For every $s > 0$, on both the periodic Hölder–Zygmund space Λ^s and its little subspace Λ_0^s ,

$$\sigma_{\text{ess}}(\mathcal{L}) = \overline{D}(0, 2^{-s-1}), \quad \sigma(\mathcal{L}) = \overline{D}(0, 2^{-s-1}) \cup \{1/2, -1/4\}.$$

The two spaces have different boundary eigenvectors and different boundary Jordan chains. At $s = 1$, both have the sharp residual operator rate $\Theta(n4^{-n})$, although only the big Zygmund space has actual length-two Jordan chains at $-1/4$.

Abstract

The inspected ProveIt report on integer C^r spectral disks leaves the noninteger Hölder and Zygmund spectra, the critical C^1 eigenspace, and finite reduction for more general trigonometric masks unresolved. We give a complete periodic solution for all positive Hölder–Zygmund orders, together with the noninteger interval extension. The proof uses an approximation norm in which the operator, modulo a finite Fourier space, is an exact contraction. This avoids a fractional regularity bootstrap and identifies every exterior spectral value by a finite matrix.

The argument applies to every nonnegative trigonometric Markov mask for every integer dilation $b \geq 2$. It determines the full Fredholm essential disk, the point spectrum, all exterior generalized eigenspaces, and the entire boundary Jordan structure. Each finite-matrix Jordan block on the critical circle gains one place in the big space; in the little space the finite-matrix block is unchanged. This yields exact residual power orders, including the same critical polynomial loss in both spaces. For the Rvachev operator, the critical classical C^1 generalized eigenspace is exactly two-dimensional and has no nontrivial Jordan chains. Classical spectral-radius and approximation results are credited explicitly; publication-level priority is not asserted. All new spectral arguments are proved here, with finite algebra checked separately by exact arithmetic.

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1 The question, the contribution, and the literature

1.1 A specific open target in ProveIt

The Fabius-function branch of ProveIt studies Rvachev’s up-function, Thue–Morse products, and transfer operators arising in Fourier-decay and mean-square asymptotics. The manuscript *Sharp C^r Spectral Disks for the Rvachev–Thue–Morse Transfer Operator* proves the integer C^r spectral formula and then lists several extensions [1, 2]. Its first research question concerns noninteger Hölder, little-Hölder, and Zygmund spaces. Its second asks about the critical C^1 eigenvalue $-1/4$, including its multiplicity. Its third asks for finite Fourier reduction for general trigonometric Markov masks.

This paper resolves the periodic spectral and generalized-eigenvector parts of those questions. It also proves the noninteger Hölder result on $[0, 1]$, gives exact operator-norm remainder orders, and supplies a general finite-matrix procedure. The sharp local C^1 resolvent problem, weighted endpoints, and the integer-Zygmund interval problem remain separate questions. The latter requires more than matching finitely many endpoint derivatives; section 9 explains the obstruction.

The source snapshot is pinned in appendix B. The earlier integer C^r result and its elementary three-mode eigenfunctions are background, not new claims of this paper. Every theorem needed for the new positive-order approximation-space classification is proved below.

1.2 What is classical

The operator is a familiar normalized transfer operator for the Thue–Morse g -function. Baake, Gohlke, Kesseböhmer, and Schindler identify its invariant measure and study its mixing and scaling properties [3]. Weighted dyadic Markov operators, including weights with zeros, already occur in Conze–Raugi and Hennion [4, 5]. Hennion’s unweighted example also provides a classical precedent for lacunary eigenfunctions filling a Hölder disk.

Exact essential spectral *radii* on Hölder and Zygmund spaces have a substantial prior theory. In particular, Baladi–Jiang–Lanford derive exact formulas on these scales [6]; more general Besov bounds appear in Nakano–Sakamoto [9]. Chen–Volkmer study trigonometric weights, Hölder quasi-compactness, and finite matrices for growth calculations [7]. Dutkay’s wavelet-Galerkin work is a related precedent for full continuous-space disks, but its low-pass normalization is different from the sine mask here [8]. Uniform trigonometric approximation and Jackson–Stechkin inequalities are likewise classical [10, 11].

The contribution made here is a complete and elementary resolution of the stated repository questions: an exact quotient contraction, the full spectral set rather than only its radius, a finite core for every finite Markov mask, the big/little boundary classification, the critical Jordan extension, and sharp residual power orders. The reviewed sources establish important antecedents; the present literature check does not certify that every specialization is new to the worldwide literature.

Target	Result here	Remaining limitation
Noninteger Hölder spectra	Exact spectrum, essential spectrum, and point spectrum on circle and interval	Endpoint-weighted variants remain open
Periodic Zygmund spectra	All $s > 0$, including integer orders	No integer interval-Zygmund theorem is inferred
Critical C^1 eigenspace	Exactly two-dimensional; all generalized vectors are ordinary eigenvectors	Sharp local C^1 resolvent growth is not settled
Finite trigonometric masks	Explicit core matrix, boundary Jordan theorem, and exact residual orders	Positivity and finite Fourier support are hypotheses

2 Spaces, normalization, and the principal theorem

2.1 The operator

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and use complex-valued functions. The repository normalization is

$$(\mathcal{L}f)(x) = \frac{1}{2} \left[\sin^2\left(\frac{\pi x}{2}\right) f(x/2) + \cos^2\left(\frac{\pi x}{2}\right) f((x+1)/2) \right]. \quad (2.1)$$

We first work with the Markov normalization

$$\mathcal{P} = 2\mathcal{L}, \quad \mathcal{P}1 = 1, \quad \|\mathcal{P}f\|_\infty \leq \|f\|_\infty. \quad (2.2)$$

The two branch terms interchange when x is replaced by $x+1$, so their sum is periodic. Every final spectral value for $\mathcal{P}$ is divided by two to obtain the corresponding value for $\mathcal{L}$.

2.2 Big and little regularity

Let $e_m(x) = \exp(2\pi imx)$ and

$$\mathcal{V}_N = \text{span}\{e_m : |m| \leq N\}, \quad E_N(f) = \inf_{p \in \mathcal{V}_N} \|f - p\|_\infty \quad (N \geq 0). \quad (2.3)$$

For $s > 0$, define the periodic Hölder–Zygmund space by the equivalent approximation norm

$$\Lambda^s = \left\{ f \in C(\mathbb{T}) : \|f\|_{\Lambda^s} := \|f\|_\infty + \sup_{N \geq 1} N^s E_N(f) < \infty \right\}. \quad (2.4)$$

Its little subspace is

$$\Lambda_0^s = \{f \in \Lambda^s : N^s E_N(f) \rightarrow 0\}. \quad (2.5)$$

It is precisely the closure of trigonometric polynomials, equivalently of smooth periodic functions, in Λ^s .

For $s = r + \alpha$, $r \geq 0$ an integer and $0 < \alpha < 1$, these are the usual spaces $C^{r,\alpha}(\mathbb{T})$ and $h^{r,\alpha}(\mathbb{T})$, with equivalent norms. At $s = m \in \mathbb{N}$, Λ^m is the Zygmund space $B_{\infty,\infty}^m$, and Λ_0^m is its little subspace. In particular, Λ^1 is defined equivalently by bounded symmetric second differences divided by $|h|$. It is not the Lipschitz space and not C^1 . One has

$$C^m(\mathbb{T}) \subseteq \Lambda_0^m(\mathbb{T}) \subseteq \Lambda^m(\mathbb{T}). \quad (2.6)$$

The approximation characterizations and these identifications are proved in appendix A; they are not additional spectral assumptions.

2.3 Spectral terminology

We use the Fredholm essential spectrum

$$\sigma_{\text{ess}}(T) = \{z \in \mathbb{C} : T - zI \text{ is not Fredholm}\}.$$

The point spectrum is $\sigma_p(T) = \{z : \ker(T - zI) \neq 0\}$. Write

$$\mathcal{G}_z(T) = \bigcup_{k \geq 1} \ker(T - zI)^k \quad (2.7)$$

for the algebraic generalized eigenspace. At a nonisolated spectral point we speak directly about this space and its Jordan chains; we do not assign a Riesz projection there.

Theorem 2.1 (Complete periodic spectral classification). *For every $s > 0$, put $\rho_s = 2^{-s-1}$. On both Λ^s and Λ_0^s ,*

$$\sigma_{\text{ess}}(\mathcal{L}) = \overline{D}(0, \rho_s), \quad \sigma(\mathcal{L}) = \overline{D}(0, \rho_s) \cup \{1/2, -1/4\}. \quad (2.8)$$

The point spectra are

$$\sigma_p(\mathcal{L}; \Lambda^s) = \overline{D}(0, \rho_s) \cup \{1/2, -1/4\}, \quad (2.9)$$

$$\sigma_p(\mathcal{L}; \Lambda_0^s) = D(0, \rho_s) \cup \{1/2, -1/4\}. \quad (2.10)$$

Every point of the closed disk in the big space, and of the open disk in the little space, has infinite geometric multiplicity.

The eigenvalue $1/2$ is isolated and simple for all $s > 0$. The eigenvalue $-1/4$ is isolated exactly when $s > 1$; there it is semisimple with two-dimensional eigenspace

$$H = \text{span}\{\sin(2\pi x), \cos(2\pi x) + 1/3\}. \quad (2.11)$$

Every exterior generalized eigenvector belongs to $F_1 = \text{span}\{e_{-1}, 1, e_1\}$.

Theorem 2.2 (The critical Jordan transition). *Let $N = \mathcal{L} + \frac{1}{4}I$. On the big Zygmund space Λ^1 ,*

$$\mathcal{G}_{-1/4}(\mathcal{L}) = \ker N^2, \quad \dim(\ker N^2 / \ker N) = 2, \quad (2.12)$$

while $\ker N$ is infinite-dimensional. There are two independent length-two Jordan chains and no chain of length three. On Λ_0^1 , and also on classical $C^1(\mathbb{T})$,

$$\mathcal{G}_{-1/4}(\mathcal{L}) = \ker N = H. \quad (2.13)$$

All other boundary generalized eigenvectors in a big Λ^s are ordinary eigenvectors. In a little space, the only possible boundary generalized vectors are the finite-core ones; for noninteger s there are none.

The same noninteger Hölder statements hold on $[0, 1]$. The critical C^1 statement also holds on $[0, 1]$. Integer Zygmund statements in this paper are periodic statements.

3 An exact finite Fourier reduction for every finite Markov mask

3.1 The general operator

Fix an integer $b \geq 2$ and a real, nonnegative trigonometric polynomial

$$a(x) = \sum_{|j| \leq d} a_j e_j(x) \quad \text{such that} \quad \frac{1}{b} \sum_{\ell=0}^{b-1} a((x + \ell)/b) = 1. \quad (3.1)$$

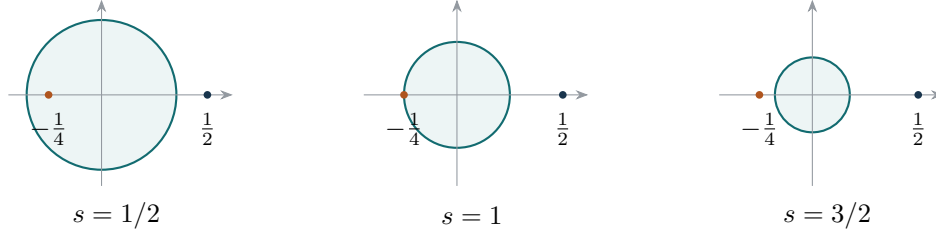


Figure 1: The spectrum of $\mathcal{L}$ as regularity increases. The essential disk has radius 2^{-s-1} . The same spectral set occurs in the big and little spaces, but their boundary eigenspaces differ. At $s = 1$, $-1/4$ meets the essential circle and the big Zygmund Jordan extension appears.

We set $a_j = 0$ for $|j| > d$. The constant mask has $d = 0$; otherwise d can be its exact degree. Define

$$(\mathcal{T}_a f)(x) = \frac{1}{b} \sum_{\ell=0}^{b-1} a((x+\ell)/b) f((x+\ell)/b), \quad (\mathcal{U}_b f)(x) = f(bx). \quad (3.2)$$

The Markov condition is equivalent to $a_0 = 1$ and $a_{bj} = 0$ for every nonzero integer j . Nonnegativity and the normalization imply

$$\|\mathcal{T}_a f\|_\infty \leq \|f\|_\infty, \quad \mathcal{T}_a \mathcal{U}_b = I. \quad (3.3)$$

For the Rvachev operator, $b = 2$ and $a(x) = 1 - \cos(2\pi x)$, so $\mathcal{T}_a = \mathcal{P}$.

Lemma 3.1 (Fourier rule and invariant core). *For each integer m ,*

$$\mathcal{T}_a e_m = \sum_{\substack{|j| \leq d \\ b|m+j}} a_j e_{(m+j)/b}, \quad \mathcal{T}_a \mathcal{V}_N \subseteq \mathcal{V}_{\lfloor (N+d)/b \rfloor}. \quad (3.4)$$

Put

$$\xi = \frac{d}{b-1}, \quad K = \lfloor \xi \rfloor, \quad F = \mathcal{V}_K. \quad (3.5)$$

Then F is invariant, and the matrix of $A = \mathcal{T}_a|_F$ in the Fourier basis indexed by $-K, \dots, K$ is

$$A_{k,m} = a_{bk-m}. \quad (3.6)$$

Proof. Insert a and e_m into (3.2). The finite sum $b^{-1} \sum_{\ell=0}^{b-1} \exp(2\pi i(m+j)\ell/b)$ is one when $b \mid m+j$ and zero otherwise. This proves the Fourier rule and the degree bound. Write $d = (b-1)K + r$ with $0 \leq r \leq b-2$; then $\lfloor (K+d)/b \rfloor = K$. The coefficient at e_k in the image of e_m is a_{bk-m} . $\square$

3.2 The shifted approximation norm

For an integer $N \geq K$ set

$$v_N = N + 1 - \xi > 0, \quad q_s(f) = \sup_{N \geq K} v_N^s E_N(f). \quad (3.7)$$

Since $v_N \asymp N$ for large N , the norm $\|f\|_\infty + q_s(f)$ defines the same Λ^s as (2.4). The little condition is equivalently $v_N^s E_N(f) \rightarrow 0$.

The seminorm q_s vanishes precisely on F . It is unchanged by adding an element of F , and it gives a complete norm on the quotient Λ^s/F . Indeed, the quotient norm induced by $\|f\|_\infty + q_s(f)$ equals

$$E_K(f) + q_s(f), \quad E_K(f) \leq v_K^{-s} q_s(f), \quad (3.8)$$

so the two quotient norms are equivalent. The same reasoning applies to the little space.

Theorem 3.2 (Exact contraction after removal of the finite core). *For $N \geq K$ and $n \geq 0$, define the integer*

$$\nu_n(N) = b^n(N+1) - d \sum_{j=0}^{n-1} b^j - 1. \quad (3.9)$$

Then

$$v_{\nu_n(N)} = b^n v_N, \quad \mathcal{T}_a^n \mathcal{V}_{\nu_n(N)} \subseteq \mathcal{V}_N, \quad (3.10)$$

$$E_N(\mathcal{T}_a^n f) \leq E_{\nu_n(N)}(f), \quad q_s(\mathcal{T}_a^n f) \leq b^{-ns} q_s(f). \quad (3.11)$$

Thus the operator Q induced on either quotient by F satisfies $\|Q^n\| \leq b^{-ns}$ in the norm q_s . For every fixed $f \in \Lambda_0^s$ one has the stronger statement

$$b^{ns} q_s(\mathcal{T}_a^n f) \rightarrow 0. \quad (3.12)$$

Proof. The affine map $\nu_1(N) = b(N+1) - d - 1$ has $\nu_1(N) \geq N$ for $N \geq K$, because its minimum increment is $b - 1 - r \geq 1$ in the notation of lemma 3.1. Also

$$\left\lfloor \frac{\nu_1(N) + d}{b} \right\rfloor = \left\lfloor N + 1 - \frac{1}{b} \right\rfloor = N.$$

Iteration gives (3.9) and the degree inclusion. The identity for v follows from $d = (b-1)\xi$.

Choose a polynomial p of degree at most $\nu_n(N)$. Its image has degree at most N , and the Markov contraction gives $E_N(\mathcal{T}_a^n f) \leq \|\mathcal{T}_a^n(f-p)\|_\infty \leq \|f-p\|_\infty$. Taking the infimum proves the first inequality in (3.11); multiplying by v_N^s and taking the supremum proves the second. Finally,

$$b^{ns} q_s(\mathcal{T}_a^n f) \leq \sup_{M \geq \nu_n(K)} v_M^s E_M(f),$$

and $\nu_n(K) \rightarrow \infty$. The right side tends to zero in the little space. $\square$

For the sine mask, $\xi = K = 1$, $v_N = N$, and $\nu_n(N) = 2^n N$. The entire mechanism then reduces to

$$\boxed{E_N(\mathcal{P}^n f) \leq E_{2^n N}(f)}. \quad (3.13)$$

This is why a finite Fourier calculation can determine all exterior spectral values without assuming that their eigenvectors are analytic or bootstrapping them through fractional derivatives.

4 The full disk and the finite exterior spectrum

4.1 Polynomial defects and lacunary series

The right inverse in (3.3) makes an infinite-dimensional defect space explicit. For every integer m not divisible by b , put

$$h_m = e_m - \mathcal{U}_b \mathcal{T}_a e_m. \quad (4.1)$$

Then h_m is a trigonometric polynomial and $\mathcal{T}_a h_m = 0$. All frequencies of $\mathcal{U}_b \mathcal{T}_a e_m$ are divisible by b . The coefficient of h_m at a nonmultiple m' of b is therefore $\delta_{m,m'}$. Thus these defects are linearly independent.

For a polynomial h and $|z| < 1$, define the uniformly convergent series

$$W_z(h) = \sum_{n=0}^{\infty} z^n \mathcal{U}_b^n h. \quad (4.2)$$

We shall use it both for kernel defects and for finite-core vectors.

Lemma 4.1 (Regularity and the exact series identity). *Let $s > 0$ and $\rho = b^{-s}$. If h is a polynomial, then $W_z(h) \in \Lambda^s$ for $|z| = \rho$, and $W_z(h) \in \Lambda_0^s$ for $|z| < \rho$. For every $|z| < 1$,*

$$(\mathcal{T}_a - zI)W_z(h) = \mathcal{T}_a h, \quad (I - z\mathcal{U}_b)W_z(h) = h. \quad (4.3)$$

At $|z| = \rho$ the partial sums are uniformly bounded in Λ^s .

Proof. Suppose first that h has degree $D \geq 1$. A partial sum through index j has degree at most Db^j . For $N \geq D$, let j be the largest integer with $Db^j \leq N$. The remaining uniform tail is at most

$$\|h\|_\infty \frac{|z|^{j+1}}{1 - |z|}.$$

When $|z| = b^{-s}$, the inequality $N < Db^{j+1}$ bounds this tail by $C_{s,h}N^{-s}$. The same estimate, with an earlier stopping point when necessary, applies uniformly to every finite partial sum. Constant polynomials cause no difficulty because their entire series is constant.

If $0 < |z| < b^{-s}$, choose $t > s$ with $|z| < b^{-t}$. The same argument, or absolute convergence in the Λ^s norm, gives membership in Λ_0^s . More explicitly, $\|\mathcal{U}_b^n h\|_{\Lambda^s} \leq C_{s,h}b^{ns}$; the series converges in that norm and each term is smooth. The case $z = 0$ is immediate.

Both identities follow by uniformly convergent summation and $\mathcal{T}_a \mathcal{U}_b^n = \mathcal{U}_b^{n-1}$ for $n \geq 1$. $\square$

Remark 4.2 (The topology of boundary convergence). The boundary series is proved to converge uniformly and to belong to the big space. It is generally not a convergent series in the big Hölder–Zygmund norm. Confusing these two assertions would incorrectly put the boundary eigenfunctions in the little space.

Theorem 4.3 (General finite-mask spectrum). *Under (3.1), let A be (3.6) and $\rho = b^{-s}$, $s > 0$. On both Λ^s and Λ_0^s ,*

$$\sigma_{\text{ess}}(\mathcal{T}_a) = \overline{D}(0, \rho), \quad \sigma(\mathcal{T}_a) = \overline{D}(0, \rho) \cup \sigma(A). \quad (4.4)$$

Furthermore,

$$\sigma_p(\mathcal{T}_a; \Lambda^s) = \overline{D}(0, \rho) \cup \sigma(A), \quad (4.5)$$

$$\sigma_p(\mathcal{T}_a; \Lambda_0^s) = D(0, \rho) \cup \sigma(A). \quad (4.6)$$

Every point of the closed disk in the big space, and of the open disk in the little space, has infinite geometric multiplicity. For $|z| > \rho$, on either space,

$$\mathcal{G}_z(\mathcal{T}_a) = \mathcal{G}_z(A) \subseteq F. \quad (4.7)$$

The same root-space equality holds at $|z| = \rho$ on the little space.

Proof. Choose a bounded linear complement to the finite-dimensional invariant space F . The quotient norm q_s identifies the operator with a bounded upper triangular block operator

$$\mathcal{T}_a = \begin{pmatrix} A & B \\ 0 & Q \end{pmatrix}, \quad \|Q^n\| \leq \rho^n. \quad (4.8)$$

Such a complement exists because a finite-dimensional subspace of a Banach space is complemented. Its second block is identified with the quotient, so the displayed power bound is valid in an equivalent norm.

For $|z| > \rho$, $Q - zI$ is invertible by its Neumann series. Elementary block inversion then shows that $\mathcal{T}_a - zI$ is invertible if and only if $A - zI$ is invertible. Projecting a generalized eigenvector to the quotient also proves (4.7). Even when $A - zI$ is singular, the full block operator is Fredholm

of index zero: multiply by invertible triangular factors to remove B , leaving a finite matrix and an invertible infinite block.

For every polynomial kernel defect h , lemma 4.1 gives $\mathcal{T}_a W_z(h) = z W_z(h)$ at every $|z| \leq \rho$ in the big space and every $|z| < \rho$ in the little space. The second identity in (4.3) makes $h \mapsto W_z(h)$ injective. Thus all interior points have infinite-dimensional kernels, so they are not Fredholm. The set of non-Fredholm operators is closed; hence the whole closed disk is essential. The exterior Fredholm statement proves that there are no further essential points. Every eigenvalue of A has a polynomial eigenvector in both spaces. This proves the spectral sets and the big-space point spectrum.

It remains to exclude extra little-space boundary eigenvectors and generalized vectors. Let f be generalized at $|z| = \rho$, and write y for its quotient class. If $y \neq 0$, let k be its Jordan height, so $(Q - zI)^k y = 0$ and $(Q - zI)^{k-1} y \neq 0$. Then

$$z^{-n} Q^n y = \sum_{j=0}^{k-1} \binom{n}{j} z^{-j} (Q - zI)^j y. \quad (4.9)$$

The right side is a nonzero vector-valued polynomial, of degree $k-1$. It cannot tend to zero. But (3.12) says precisely that its norm tends to zero in the little quotient. Thus $y = 0$ and $f \in F$. This proves the boundary root-space statement and completes (4.6). $\square$

4.2 Two useful consequences

For each compact operator C on either regularity space and every $n \geq 1$,

$$\|\mathcal{T}_a^n - C\| \geq \rho^n. \quad (4.10)$$

Indeed, the spectrum of the Calkin image of $\mathcal{T}_a$ is the Fredholm essential disk. Spectral mapping puts a disk of radius ρ^n in the spectrum of its n th power, and the quotient norm dominates spectral radius. Equivalently, (4.10) is the unavoidable exponential obstruction to approximation by compact operators.

There is also a precise regularity obstruction. If $\mathcal{T}_a f = z f$, $f \notin F$, and f is continuous, then exact degree collapse and Markov contraction give

$$E_{\nu_n(K)}(f) \geq |z|^n E_K(f). \quad (4.11)$$

Thus a noncore eigenvector at $|z| = b^{-s}$ cannot have little order s , or any higher Hölder–Zygmund regularity. For a quotient Jordan chain of height k , the same argument and (4.9) give the lower bound $cn^{k-1}|z|^n$ along this approximation sequence. A longer quotient chain would require a logarithmic loss in approximation order.

5 Every boundary Jordan chain is determined by the finite core

Theorem 5.1 (Boundary extension by one Jordan level). *Assume (3.1), fix $s > 0$, and let $|z| = \rho = b^{-s}$. On the big space Λ^s , put $N = \mathcal{T}_a - zI$. For every $j \geq 1$, the map induced by N is an isomorphism*

$$\frac{\ker N^{j+1}}{\ker N} \xrightarrow{\cong} \ker(A - zI)^j. \quad (5.1)$$

In particular,

$$\frac{\mathcal{G}_z(\mathcal{T}_a)}{\ker(\mathcal{T}_a - zI)} \cong \mathcal{G}_z(A). \quad (5.2)$$

If z is not an eigenvalue of A , all its generalized vectors are ordinary eigenvectors. If the largest Jordan block of A at z has length $\ell \geq 1$, the largest Jordan chain in the big space has length

exactly $\ell + 1$. Each finite-core block gains one place; the remaining boundary eigenvectors have no further generalized level.

On the little space, $\mathcal{G}_z(\mathcal{T}_a) = \mathcal{G}_z(A)$, so all finite-core Jordan lengths are unchanged.

Proof. Let f be any generalized vector in the big space and project it to the quotient. Formula (4.9), together with $\|Q^n\| \leq \rho^n$, shows that the quotient Jordan height is at most one. Therefore $Nf \in F$. It is itself generalized at z , so $Nf \in \mathcal{G}_z(A)$. For $f \in \ker N^{j+1}$ this gives $Nf \in \ker(A - zI)^j$. The kernel of the induced map is exactly $\ker N$.

To prove surjectivity, take $h \in \mathcal{G}_z(A)$. Since $z \neq 0$, A is invertible on this finite generalized eigenspace. Set

$$v = (A|_{\mathcal{G}_z(A)})^{-1}h, \quad R_z h = W_z(v) = \sum_{n=0}^{\infty} z^n \mathcal{U}_b^n v. \quad (5.3)$$

The vector v is a polynomial. By lemma 4.1, $R_z h$ belongs to the big space and

$$(\mathcal{T}_a - zI)R_z h = \mathcal{T}_a v = Av = h. \quad (5.4)$$

If $(A - zI)^j h = 0$, then $N^{j+1}R_z h = 0$. This proves surjectivity in (5.1) and hence all quotient identities.

For the maximal length, $Nf \in \mathcal{G}_z(A)$ implies $N^{\ell+1}f = 0$ for every generalized f . Choosing h of core Jordan height ℓ in (5.3) produces a vector of height $\ell + 1$. More precisely, the dimensions in (5.1) are the sums of $\min(j, m)$ over the core block lengths m , which are exactly the dimensions obtained by increasing each such block length by one. The eigenspace itself is infinite-dimensional. The little-space assertion was established in theorem 4.3. $\square$

The theorem describes an embedded boundary phenomenon. Its statement does not depend on isolation of the spectral point, and it does not introduce a spectral projection on the essential circle.

Proposition 5.2 (Interior chains of arbitrary finite length). *At each $|z| < b^{-s}$, both spaces contain Jordan chains of every finite length, with infinitely many independent bottom eigenvectors.*

Proof. For a polynomial kernel seed h , the series $W_z(h)$ is norm-holomorphic in $|z| < b^{-s}$, because its differentiated terms are bounded by a polynomial in n times $(|z|b^s)^n$. Differentiating $\mathcal{T}_a W_z(h) = zW_z(h)$ gives

$$(\mathcal{T}_a - zI) \frac{W_z^{(j)}(h)}{j!} = \frac{W_z^{(j-1)}(h)}{(j-1)!} \quad (j \geq 1).$$

The bottom vector is nonzero and is injective in the seed. At $z = 0$, the chains are simply $h, \mathcal{U}_b h, \mathcal{U}_b^2 h, \dots$ $\square$

6 The Rvachev matrix and the critical classical C^1 question

6.1 Three Fourier modes

For $a(x) = 1 - \cos(2\pi x)$, (3.4) is

$$\mathcal{P}e_{2k} = e_k, \quad \mathcal{P}e_{2k+1} = -\frac{1}{2}(e_k + e_{k+1}). \quad (6.1)$$

In the ordered basis $(e_{-1}, 1, e_1)$,

$$A = \begin{pmatrix} -1/2 & 0 & 0 \\ -1/2 & 1 & -1/2 \\ 0 & 0 & -1/2 \end{pmatrix}, \quad \det(zI - A) = (z-1)(z+1/2)^2. \quad (6.2)$$

The matrix is diagonalizable. Its 1-eigenspace consists of constants, and its $-1/2$ -eigenspace is H in (2.11). Equivalently, it is spanned by $e_{-1} + 1/3$ and $e_1 + 1/3$. Theorem 4.3 now proves theorem 2.1 after division of all eigenvalues by two.

For explicit boundary eigenfunctions, one may use the independent seeds

$$h_j = e_{2j+1} + \frac{1}{2}e_{2j} + \frac{1}{2}e_{2j+2}, \quad j \geq 1. \quad (6.3)$$

They satisfy $\mathcal{P}h_j = 0$. For $|z| = 2^{-s}$,

$$f_{z,j}(x) = \sum_{n=0}^{\infty} z^n h_j(2^n x) \quad (6.4)$$

belongs to the big Λ^s , has $\mathcal{P}$ -eigenvalue z , and gives an $\mathcal{L}$ -eigenfunction at $z/2$. Their sharp approximation obstruction is (4.11). This explains, constructively, why the common essential circle has very different point spectra in the two spaces.

6.2 Explicit critical Jordan partners

Put $z = -1/2$ and let $h \in H$. Since $\mathcal{P}h = zh$, the function

$$F_h(x) = \sum_{n=0}^{\infty} (-1/2)^n h(2^n x) \quad (6.5)$$

belongs to Λ^1 and satisfies

$$(\mathcal{P} + \frac{1}{2}I)F_h = -\frac{1}{2}h, \quad (\mathcal{L} + \frac{1}{4}I)F_h = -\frac{1}{4}h. \quad (6.6)$$

Thus $-4F_h$ is a Jordan partner for h in the repository normalization. For example, taking $h = e_1 + 1/3$ gives the concrete complex function

$$F_h(x) = \frac{2}{9} + \sum_{n=0}^{\infty} (-1/2)^n \exp(2\pi i 2^n x). \quad (6.7)$$

Real and imaginary parts give real-valued partners.

The core eigenvalue $-1/2$ has two blocks of length one. Theorem 5.1 therefore proves that the full big-space root space has exactly two additional generalized directions and no third level. This proves (2.12). Neither nonzero partner lies in Λ_0^1 : little-space boundary rigidity would put it in the diagonalizable finite core, contradicting (6.6).

6.3 Why the answer on C^1 is different

For $f \in C^1(\mathbb{T})$,

$$\|f(\cdot + 2h) - 2f(\cdot + h) + f\|_{\infty} \leq |h| \omega(f', |h|) = o(|h|), \quad (6.8)$$

where ω is the uniform modulus of continuity. The approximation lemma in appendix A implies $E_N(f) = o(N^{-1})$. Hence $C^1 \subseteq \Lambda_0^1$, and any C^1 generalized vector at $-1/2$ must be in the finite core. All core vectors are smooth, so the generalized eigenspace on C^1 is exactly H .

More generally, for every integer $r \geq 1$, every classical C^r generalized vector at $|z| = 2^{-r}$ belongs to F_1 . At $r \geq 2$ there are no such vectors because neither 1 nor $-1/2$ lies on that circle. This is a boundary statement; it does not identify the classical C^r norm with a Zygmund norm.

Space	Spectral set	Boundary eigenspace at $-1/4$ when $s = 1$	Nontrivial boundary Jordan chains
Big Λ^s	Disk plus $1/2, -1/4$	Infinite-dimensional	Two length-two blocks at $s = 1$
Little Λ_0^s	The same set	Exactly H , dimension two	None for the sine mask
Classical C^1	Integer formula from [1]	Exactly H , including all generalized vectors	None

7 Sharp operator powers, including the critical polynomial loss

7.1 A general exact-order theorem

The finite-core reduction gives more than spectral radii. It determines the exact polynomial factor multiplying the residual exponential rate. For a sequence of positive numbers, $u_n = \Theta(v_n)$ means that $cv_n \leq u_n \leq Cv_n$ for all sufficiently large n , with constants $0 < c \leq C < \infty$.

Theorem 7.1 (Exact residual power order). *Let $\mathcal{T}_a$, A , s , and $\rho = b^{-s}$ be as in theorem 4.3. Let Π_{out} be the sum of the Riesz projections at all eigenvalues of A with modulus strictly greater than ρ , and put*

$$R_n = \mathcal{T}_a^n(I - \Pi_{\text{out}}), \quad n \geq 1. \quad (7.1)$$

If A has no eigenvalue on $|z| = \rho$, set $\ell = 0$. Otherwise let ℓ be the largest Jordan block length among its eigenvalues on that circle. On both the big and the little space,

$$\|R_n\| = \Theta(n^\ell \rho^n). \quad (7.2)$$

For every fixed f in the little space,

$$\|R_n f\| = o(n^\ell \rho^n). \quad (7.3)$$

Thus the operator-norm lower bound in the little space is attained by inputs depending on n , and is compatible with the sharper individual-vector statement.

Proof. Use the block form (4.8), and split the finite core into its exterior and remaining generalized eigenspaces. In these coordinates write A_{out} for the exterior block, A_0 for the remaining block, and B_{out}, B_0 for the two components of B . The series

$$S = \sum_{k=0}^{\infty} A_{\text{out}}^{-k-1} B_{\text{out}} Q^k \quad (7.4)$$

converges: the minimum exterior modulus exceeds ρ , so any polynomial factors from finite Jordan blocks are dominated geometrically. It satisfies $A_{\text{out}} S - S Q = B_{\text{out}}$. Replacing the exterior coordinate x by $x + S y$ removes its coupling to the quotient. The residual restriction is consequently similar to

$$T_0 = \begin{pmatrix} A_0 & B_0 \\ 0 & Q \end{pmatrix}, \quad T_0^n = \begin{pmatrix} A_0^n & \sum_{j=0}^{n-1} A_0^{n-1-j} B_0 Q^j \\ 0 & Q^n \end{pmatrix}. \quad (7.5)$$

If $\ell = 0$, there is a $\beta < \rho$ such that $\|A_0^k\| \leq C\beta^k$. The convolution in (7.5) is bounded by $C\rho^{n-1} \sum_{k \geq 0} (\beta/\rho)^k$, proving $\|R_n\| \leq C'\rho^n$. The matching lower bound is (4.10), since $\mathcal{T}_a^n \Pi_{\text{out}}$ is finite-rank.

If $\ell \geq 1$, finite-dimensional Jordan theory gives $\|A_0^k\| \leq C(k+1)^{\ell-1}\rho^k$. The convolution is now $O(n^\ell \rho^n)$, proving the upper bound. For the lower bound, choose a critical core eigenvalue z , $|z| = \rho$, and a core vector v of Jordan height ℓ . The polynomials

$$F_M = \sum_{k=0}^M z^k \mathcal{U}_b^k v \quad (7.6)$$

have uniformly bounded Λ^s norms by lemma 4.1; hence they are uniformly bounded inputs in both spaces. For $M \geq n$,

$$\mathcal{T}_a^n F_M = z^n \left[\sum_{j=1}^n (A/z)^j v + F_{M-n} \right]. \quad (7.7)$$

To see this, separate the terms $k < n$, on which $\mathcal{T}_a^n \mathcal{U}_b^k v = A^{n-k} v$, from those $k \geq n$, on which the right inverse removes n copies of $\mathcal{U}_b$.

Write $D = A - zI$ on the critical generalized eigenspace. The finite sum in brackets is a vector-valued polynomial in n of degree ℓ , with leading coefficient $D^{\ell-1}v/(\ell!z^{\ell-1}) \neq 0$. The term F_{M-n} is uniformly bounded in the uniform norm. Thus the output has norm at least $cn^\ell \rho^n$ for large n ; it suffices to choose $M = n$. Moreover, $\Pi_{\text{out}} F_M = 0$. Indeed, for fixed M all sufficiently late iterates lie in the critical generalized core, while the operator is invertible on the exterior generalized space. Therefore the output just computed is also $R_n F_M$. This proves the lower bound on both spaces.

Finally, on the little quotient, $\rho^{-j} Q^j y \rightarrow 0$ for every fixed y . If $\ell = 0$, insert this into (7.5) and use a dominated geometric sum. If $\ell \geq 1$, divide the convolution by $n^\ell \rho^n$. Finitely many early indices tend to zero, and the remaining terms are bounded by an arbitrarily small constant times the normalized sum of $(n-j)^{\ell-1}$. The finite block itself is $O(n^{\ell-1} \rho^n)$. This proves (7.3). $\square$

The lower bound at a critical resonance concerns the stated residual after removal of the isolated exterior modes. It does not assert that the factor n^ℓ survives every possible n -dependent compact correction. The universal obstruction against arbitrary compact corrections is the exponential bound (4.10).

7.2 Exact Rvachev rates

Let Π_+ be the rank-one Riesz projection of $\mathcal{L}$ at $1/2$. For $s > 1$, let Π_- be its rank-two Riesz projection at $-1/4$. Then theorem 7.1, after multiplication by 2^{-n} , gives the following complete rate table on periodic spaces:

regularity	residual operator	operator-norm order
$0 < s < 1$	$\mathcal{L}^n - 2^{-n} \Pi_+$	$\Theta(2^{-(s+1)n})$
$s = 1$	$\mathcal{L}^n - 2^{-n} \Pi_+$	$\Theta(n4^{-n})$
$s > 1$	$\mathcal{L}^n - 2^{-n} \Pi_+ - (-1/4)^n \Pi_-$	$\Theta(2^{-(s+1)n})$

(7.8)

Both Λ^s and Λ_0^s obey this table. There is no ε loss in the exponential base.

At $s = 1$, the elementary special case of (7.7) is especially informative. If $z = -1/2$, $h \in H$, and $F_M = \sum_{k=0}^M z^k \mathcal{U}_2^k h$, then

$$\mathcal{P}^n F_M = z^n (nh + F_{M-n}), \quad M \geq n. \quad (7.9)$$

The inputs are smooth and uniformly bounded in the Zygmund norm. They therefore produce the factor n even in the little Zygmund operator norm, where the infinite Jordan partner is absent. They need not be uniformly bounded in the C^1 norm. Accordingly, (7.8) is not asserted as a critical C^1 rate theorem.

8 A finite mask with genuinely fractional resonance thresholds

Consider

$$b = 2, \quad a(x) = 1 - \cos(6\pi x). \quad (8.1)$$

It is nonnegative and satisfies the Markov condition. Here $d = 3$, $K = 3$, and the core is seven-dimensional. The Fourier rule is

$$\mathcal{T}_a e_{2k} = e_k, \quad \mathcal{T}_a e_m = -\frac{1}{2}(e_{(m-3)/2} + e_{(m+3)/2}) \quad (m \text{ odd}).$$

In the basis indexed by $-3, -2, \dots, 3$, the matrix is

$$A_3 = \begin{pmatrix} -1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1/2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 & 0 \\ -1/2 & 0 & 0 & 1 & 0 & 0 & -1/2 \\ 0 & 0 & -1/2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1/2 \end{pmatrix}. \quad (8.2)$$

Exact arithmetic yields

$$\det(zI - A_3) = \frac{(z-1)(2z+1)^2(2z^2-z+1)(2z^2+z+1)}{16}. \quad (8.3)$$

Its minimal polynomial is the squarefree polynomial

$$(z-1)(z+1/2)(z^2-z/2+1/2)(z^2+z/2+1/2). \quad (8.4)$$

Thus all finite-core modes are semisimple. In addition to 1 and $-1/2$, there are four simple complex modes

$$z = \frac{1 \pm i\sqrt{7}}{4}, \quad z = \frac{-1 \pm i\sqrt{7}}{4}, \quad |z| = \frac{1}{\sqrt{2}}. \quad (8.5)$$

The matrix, characteristic polynomial, minimal polynomial, and ranks are verified by the included exact program. They are finite algebraic identities and can also be checked by direct multiplication.

Theorems 4.3, 5.1 and 7.1 give the complete interpretation:

- The essential disk is $\overline{D}(0, 2^{-s})$ for every $s > 0$.
- The four complex modes are outside it exactly when $s > 1/2$. At $s = 1/2$, the little-space boundary root spaces are four one-dimensional smooth eigenspaces; the other boundary points have no little-space eigenvectors.
- On big $\Lambda^{1/2}$, every boundary point has infinite geometric multiplicity, and each of the four complex core modes has one additional Jordan level. The largest chains have length two.
- After removing exterior modes, the residual order at $s = 1/2$ is $\Theta(n2^{-n/2})$ in both big and little spaces. At $s = 1$, the two $-1/2$ core modes produce another critical order $\Theta(n2^{-n})$.

This example demonstrates that the critical mechanism is not confined to an integer regularity or a negative real resonance. Its location is determined exactly by the moduli of the finite matrix eigenvalues.

9 The interval extension and the endpoint limitation

9.1 Noninteger Hölder spaces

Let $s = r + \alpha$, with $r \geq 0$ an integer and $0 < \alpha < 1$. On $X = C^{r,\alpha}([0, 1])$, or on its little subspace, define

$$\Delta_j(f) = f^{(j)}(1) - f^{(j)}(0), \quad 0 \leq j \leq r. \quad (9.1)$$

The common kernel is a closed subspace Y naturally isomorphic to the corresponding periodic space. Matching derivatives gives a C^r periodic extension. For the highest derivative, a difference across the seam is bounded by the sum of the two endpoint differences, and

$$a^\alpha + b^\alpha \leq 2^{1-\alpha}(a + b)^\alpha.$$

The same estimate with vanishing local constants preserves the little condition. Hermite interpolation realizes arbitrary jet mismatches by smooth polynomials, so Y has codimension $r + 1$.

For a smooth weight ω with $\omega(x+1) = 1 - \omega(x)$, let $P_\omega f = \omega(x)f(x/2) + (1 - \omega(x))f((x+1)/2)$. Leibniz' rule gives the exact identity

$$\Delta_j(P_\omega f) = 2^{-j}\omega(0)\Delta_j(f) + \sum_{k=1}^j \binom{j}{k} 2^{-(j-k)}\omega^{(k)}(0)\Delta_{j-k}(f). \quad (9.2)$$

To verify it, evaluate derivatives at both endpoints. Their midpoint contributions cancel, and $\omega^{(k)}(1) = -\omega^{(k)}(0)$ for $k \geq 1$ supplies the remaining signs. This algebra is also present in the earlier integer report [1]; only its application to the noninteger norms is needed here.

For the sine weight $\omega(x) = \sin^2(\pi x/2)$, $\omega(0) = 0$ and all odd derivatives at zero vanish. The quotient matrix J of $\mathcal{P}$ on the mismatch jets lowers the index by at least two. In particular,

$$J^R = 0, \quad R = \lfloor r/2 \rfloor + 1, \quad \mathcal{P}^R X \subseteq Y. \quad (9.3)$$

Theorem 9.1 (Noninteger interval classification). *For every noninteger $s = r + \alpha > 0$, the spectrum, Fredholm essential spectrum, point spectrum, exterior root spaces, and big/little boundary classification of theorems 2.1 and 2.2 hold on $C^{r,\alpha}([0, 1])$ and $h^{r,\alpha}([0, 1])$. Every nonzero generalized eigenvector is periodic through its available jets and is exactly one of the corresponding periodic generalized vectors. The noncritical power orders in (7.8) also hold on these interval spaces. On classical $C^1([0, 1])$ the critical generalized eigenspace at $-1/4$ is precisely H .*

Proof. With respect to a bounded complement of Y , the operator is triangular with the periodic restriction and the nilpotent finite quotient J . For $z \neq 0$, $J - zI$ is invertible, so block inversion identifies the interval and periodic resolvents there. Zero already belongs to the periodic essential disk. Finite-dimensional extensions preserve Fredholm essential spectrum.

If $(\mathcal{P} - zI)^m f = 0$ with $z \neq 0$, applying all mismatch functionals gives $(J - zI)^m \Delta(f) = 0$, whence $\Delta(f) = 0$. This identifies all nonzero generalized vectors. The interior zero eigenvalue still has infinite-dimensional kernel because the periodic defect functions remain available.

For power estimates, (9.3) sends every interval input to the periodic space after the fixed number R of iterates. The isolated exterior projections intertwine with this inclusion and with $\mathcal{P}^R$. Applying the periodic upper estimate after that fixed transient gives the same exponential order; restricting inputs to periodic functions gives the matching lower bound. Finally, the classical C^1 jet calculation is identical and the periodic critical root-space statement was proved in section 6. $\square$

9.2 Why this does not prove an integer interval-Zygmund theorem

It is tempting to identify an interval Zygmund function with its periodic extension whenever the available endpoint derivatives match. That identification fails already at order one. Define

$$f(x) = x \log x \quad (0 < x \leq 1), \quad f(0) = 0. \quad (9.4)$$

This belongs to the interval Zygmund class and has $f(0) = f(1) = 0$. For example, scaling $x = hy$ reduces its second difference divided by h to a bounded function of $y \geq 1$. But the symmetric second difference of its periodic extension at the seam is

$$h \log h + (1 - h) \log(1 - h),$$

whose quotient by h is unbounded as $h \downarrow 0$. Thus the finite-jet proof is insufficient for integer Zygmund spaces. The sine mask's endpoint zeros may provide another route after taking powers, but that requires a separate bounded gluing theorem. No such interval conclusion is included in the proved scope here.

10 Interpretation and a finite certificate procedure

10.1 What the norm changes

The smooth three-mode eigenfunctions do not determine the full spectrum on a finite-regularity space. The additional disk consists of oscillations that are moved down the Fourier scale by the transfer operator. A polynomial kernel seed can be repeated at frequencies b^n with amplitudes z^n ; the threshold $|z| = b^{-s}$ is exactly where those amplitudes have regularity order s .

The big space admits a bounded normalized approximation error $N^s E_N(f)$. The little space requires that error to tend to zero. This small change leaves the spectrum as a closed set unchanged, but removes every noncore boundary eigenfunction. It also removes the extra boundary Jordan level. Finite partial sums of those missing vectors remain uniformly bounded, so their absence does not remove the critical polynomial factor from the operator norm.

The finite core and the essential disk consequently encode different information. The core records exact low-frequency resonances, their algebraic multiplicities, and their threshold crossings. The quotient records the unavoidable rate of transport through the frequency scales. Their interaction at equal modulus is measured exactly by theorems 5.1 and 7.1.

10.2 A proof-oriented finite computation

For a supplied finite trigonometric mask, the analytic theorem reduces the remaining spectral work to the following finite tasks.

1. Specify b and the coefficients a_j , verify conjugate symmetry, and prove nonnegativity. For the two examples in this paper the identity $1 - \cos(2\pi dx) = |1 - e_d(x)|^2/2$ is an immediate certificate. For an arbitrary supplied mask, positivity is an input obligation, not something inferred from a numerical plot.
2. Verify $a_0 = 1$ and $a_{bj} = 0$ for all nonzero multiples of b . These are exactly the finite Markov normalization conditions.
3. Compute $K = \lfloor d/(b-1) \rfloor$ and the matrix $A_{k,m} = a_{bk-m}$ for $|k|, |m| \leq K$.
4. Compute its characteristic and minimal polynomials and the dimensions of its generalized eigenspaces. These determine the exterior resonances and their Jordan structures.

5. For each nonzero matrix eigenvalue z , record the critical order $s_z = -\log_b |z|$ when it is positive. Comparison of s with these finitely many thresholds gives the full regularity-dependent spectral classification.
6. At a critical threshold, use [theorem 5.1](#) to obtain the big-space extra Jordan level and the unchanged little-space blocks. The largest critical block length gives the residual factor n^ℓ .

The degree bound gives a guaranteed core. Special coefficient cancellations or symmetries may permit a smaller matrix; no minimal dimension for every individual mask is asserted. The exact index $\nu_n(N)$ is sharp for the formal degree recurrence $D \mapsto \lfloor (D + d)/b \rfloor$, a different statement from sharpness of the actual Fourier degree for each mask.

10.3 What has been checked computationally

The included standard-library Python program uses integers and exact rational numbers. It verifies the two core matrices, their characteristic and minimal polynomials, diagonalizability, and the displayed eigenspaces. It also checks the affine degree identities, polynomial defects, the right-inverse identity on finite characters, and the finite critical iterate formula. The result is 47,127 successful exact checks, summarized in [appendix C](#).

The analytic statements are not inferred from those finite calculations. In particular, no eigenvalue plot or finite truncation can prove that a whole disk is essential, that a boundary series belongs to a specified Banach space, or that the little-space root spaces are exhausted. Those conclusions follow from the proofs in [sections 3 to 5 and 7](#).

11 Further research questions and proposed next steps

The following questions are concrete continuations of this work. “Open here” means not settled by this article; it is not a claim that no related result exists in the literature.

Question 11.1 (The integer-Zygmund interval problem). Determine the spectrum and boundary root spaces of (2.1) on the ordinary interval Zygmund spaces. A promising first step is to prove that a sufficiently large power sends an interval function boundedly into the periodic Zygmund space. The R -fold weight contains endpoint zeros of order $2R$, suggesting the condition $2R > s$. The missing ingredient is a gluing estimate for finite differences crossing the seam, not another calculation of endpoint jets.

Question 11.2 (The classical C^1 resolvent at $-1/4$). The generalized eigenspace is now known exactly, but what is the sharp growth of the resolvent as z approaches $-1/4$ from outside the essential disk? Determine whether the C^1 norm admits an exact $O(4^{-n})$ residual bound or requires a slower prefactor. The Zygmund lower-bound polynomials are not uniformly bounded in C^1 , so they do not answer this. A useful first target is a two-sided estimate on real radial approaches, followed by angular approaches.

Question 11.3 (Logarithmic regularity at a critical circle). Replace $N^s E_N(f)$ by $N^s (\log(2 + N))^\beta E_N(f)$, and classify the boundary eigenspaces, Jordan levels, and operator powers as β varies. The explicit series and the lower bound (4.11) suggest a precise hierarchy. The shifted index formula remains exact, while the logarithmic weight introduces a controlled slowly varying ratio.

Question 11.4 (Sharp resolvent and pseudospectral constants). The quotient estimate gives $\|(zI - Q)^{-1}\| \leq (|z| - b^{-s})^{-1}$ outside the disk in the adapted norm. Combine this with the finite Sylvester transformation to obtain explicit constants for the full resolvent, especially near a critical finite-core resonance. Determine which blow-up exponents are invariant under changing to the usual Hölder norm, and which constants reflect the conditioning of the core eigenbasis.

Question 11.5 (Stability after loss of finite Fourier support). For a smooth nonnegative Markov weight with a small infinite Fourier tail, quantify the displacement of the isolated resonances of its finite truncation. One must preserve positivity and normalization while controlling the perturbation in the relevant operator norm. A constructive target is a validated enclosure for each exterior resonance together with a tail estimate that separates it from the essential disk.

Question 11.6 (Other endpoint ladders and their collisions). For a general dyadic Markov weight, the mismatch quotient has diagonal $\omega(0)2^{-j}$. When one of these values collides with a periodic finite-core resonance, classify the coupled Jordan chains. A finite matrix alone records the jet block, but the crossing block depends on how endpoint data feed the periodic part. Explicit dual resonance functionals may provide a computable obstruction and lifting criterion.

Question 11.7 (Beyond nonnegative Markov masks). Extend the quotient method to signed or complex trigonometric weights. The Fourier core and degree filtration remain exact, but the uniform contraction need not hold. Identify the correct growth factor replacing one in (3.3), and determine when the essential spectrum remains a disk rather than merely lying in one. This would separate the algebraic part of the method from positivity.

Question 11.8 (Anisotropic toral expansion). Replace $x \mapsto bx$ by an expanding integer matrix on a higher-dimensional torus. Find a finite frequency region absorbing all sufficiently small descendants, and choose an approximation norm adapted to the different expansion directions. The scalar shifted index suggests using expanding convex frequency regions. The main challenge is the interaction of anisotropic regularity with the essential spectral set.

Question 11.9 (Optimal compact corrections at criticality). The isolated-mode residual has order $n^\ell \rho^n$, while the universal lower bound against arbitrary compact corrections is only ρ^n . Determine the exact approximation numbers of $\mathcal{T}_a^n$ and the minimum rank needed to suppress the critical polynomial factor. This asks for a quantitative distinction between finite resonance removal and more general n -dependent approximation.

Question 11.10 (A verified spectral certificate interface). Formalize the finite Fourier rule, shifted indices, quotient norm, polynomial defect series, and the root-space exact sequence in a proof assistant. A certificate for the finite matrix should be accepted only after positivity and normalization are proved. The expensive reusable infrastructure is the approximation-space equivalence and Banach quotient theory; once those are available, the mask-specific spectral proof reduces to rational or algebraic matrix calculations.

A Approximation spaces and their classical regularity meanings

This appendix supplies the analytic identification used throughout. The ideas are classical Jackson approximation and Bernstein estimates; [10, 11] give broader treatments. We include the proof to make the exact big/little convention and the integer endpoints explicit.

For an integer $k \geq 1$, put

$$\Delta_h^k f(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x + jh). \quad (\text{A.1})$$

All arguments are on periodic lifts, and $0 < |h| \leq 1/2$ suffices.

Proposition A.1 (Approximation characterization). *Let $s > 0$ and choose any integer $k > s$. The norm*

$$\|f\|_\infty + \sup_{0 < |h| \leq 1/2} |h|^{-s} \|\Delta_h^k f\|_\infty \quad (\text{A.2})$$

is equivalent to (2.4). The condition $N^s E_N(f) \rightarrow 0$ is equivalent both to polynomial density in this norm and to

$$\left\| \Delta_h^k f \right\|_\infty = o(|h|^s) \quad (h \rightarrow 0). \quad (\text{A.3})$$

For noninteger $s = r + \alpha$, these norms identify the space with $C^{r,\alpha}$ and its little subspace. For integer $s = m$, they give the usual Zygmund space and its little subspace.

A.1 Direct approximation by a Jackson kernel

Choose an integer p with $2p > s + 1$. There are normalized nonnegative trigonometric kernels J_N of degree at most N satisfying, on $[-1/2, 1/2]$,

$$\int_{-1/2}^{1/2} J_N(t) dt = 1, \quad J_N(t) \leq C_p N (1 + N|t|)^{-2p}. \quad (\text{A.4})$$

For completeness, for N large take an integer $L = \lfloor N/p \rfloor + 1$ and normalize $(\sin(\pi Lt)/\sin(\pi t))^{2p}$. Its degree is $p(L-1) \leq N$. The normalizing integral is comparable to L^{2p-1} : a neighborhood of size $1/L$ gives the lower bound, and the elementary bound $|\sin(\pi Lt)/\sin(\pi t)| \leq C \min(L, |t|^{-1})$ gives the upper bound. These estimates imply (A.4); the finitely many small N are absorbed by adjusting the constant. In particular,

$$\int J_N(t) |t|^s dt \leq C_{p,s} N^{-s}. \quad (\text{A.5})$$

Define

$$p_N(x) = \int_{-1/2}^{1/2} J_N(t) \sum_{j=1}^k (-1)^{j+1} \binom{k}{j} f(x + jt) dt. \quad (\text{A.6})$$

This is a trigonometric polynomial of degree at most N . Indeed, its Fourier coefficient at index ℓ is the coefficient of f there times a combination of $\hat{J}_N(-j\ell)$; every such term vanishes if $|\ell| > N$. Normalization gives

$$f - p_N = (-1)^k \int J_N(t) \Delta_t^k f dt. \quad (\text{A.7})$$

If (A.2) is bounded, the moment estimate proves $E_N(f) \leq CN^{-s}$. If (A.3) holds, split the integral at a fixed small δ . The part $|t| < \delta$ is bounded by εCN^{-s} ; after multiplication by N^s , the remaining part tends to zero because $2p > s + 1$. This proves the little-o direct statement.

A.2 The inverse estimate

Choose best approximants $p_j \in \mathcal{V}_{2^j}$. They exist because the polynomial space is finite-dimensional. Set $q_0 = p_0$ and $q_j = p_j - p_{j-1}$ for $j \geq 1$. If $M = \|f\|_\infty + \sup_N N^s E_N(f)$, then

$$f = \sum_{j=0}^{\infty} q_j \quad \text{uniformly,} \quad \|q_j\|_\infty \leq C_s M 2^{-js} \quad (j \geq 1). \quad (\text{A.8})$$

The usual Bernstein estimate for trigonometric polynomials, together with the trivial difference bound, gives

$$\left\| \Delta_h^k q_j \right\|_\infty \leq C_k \min\{1, (2^j |h|)^k\} \|q_j\|_\infty. \quad (\text{A.9})$$

Only a constant depending on k is needed here. One elementary way to obtain the derivative bound is to reproduce a polynomial of degree N by convolution with a smooth Fourier cutoff equal to one on $[-N, N]$. The k th derivative of that kernel has L^1 norm $O_k(N^k)$, which gives $\|q^{(k)}\|_\infty \leq C_k N^k \|q\|_\infty$; repeated integration then gives (A.9).

Choose J so that $2^J|h| \leq 1 < 2^{J+1}|h|$. For $j \leq J$, sum the bound $CM|h|^k 2^{j(k-s)}$; for $j > J$, sum $CM2^{-js}$. Both geometric sums are $O(M|h|^s)$ because $k > s$. This proves the inverse norm estimate. If $N^s E_N(f) \rightarrow 0$, the normalized coefficients $2^{js} \|q_j\|_\infty$ tend to zero. A finite initial sum is smooth and has difference $O(|h|^k) = o(|h|^s)$; the remaining sum has an arbitrarily small constant in the preceding estimate. This proves (A.3).

A.3 Density, completeness, and noninteger derivatives

The little condition is exactly polynomial density. One direction is immediate: for a fixed polynomial p and all $N \geq \deg p$, $E_N(f) = E_N(f - p)$. Conversely, if $N^s E_N(f) \rightarrow 0$ and p_N is a best degree- N approximant, then

$$\sup_{1 \leq M < N} M^s E_M(f - p_N) \leq N^s E_N(f), \quad \sup_{M \geq N} M^s E_M(f - p_N) = \sup_{M \geq N} M^s E_M(f).$$

Both sides tend to zero, as does $\|f - p_N\|_\infty$. Smooth periodic functions have superalgebraically small approximation errors, for instance by repeated integration by parts of their Fourier coefficients. Thus closure of smooth functions and closure of polynomials coincide.

Completeness follows directly. A Cauchy sequence in the approximation norm has a uniform limit f . Each E_N is 1-Lipschitz in the uniform norm. Passing to the limit in the uniform weighted bounds on $E_N(f_j - f_k)$ proves convergence in the full approximation norm. The little space is closed by its density characterization.

If $s = r + \alpha$ with $0 < \alpha < 1$, apply the derivative Bernstein bound to (A.8). The series of derivatives through order r converges uniformly because $\sum_j 2^{jr} 2^{-js} = \sum_j 2^{-j\alpha} < \infty$. Its sum is the corresponding derivative of f . For increments of the r th derivative, the low-frequency estimate is $C|h|2^{j(1-\alpha)}$ and the high-frequency estimate is $C2^{-j\alpha}$. Splitting at J gives the α -Hölder bound. Conversely,

$$\left\| \Delta_h^{r+1} f \right\|_\infty \leq |h|^r \omega(f^{(r)}, |h|) \quad (\text{A.10})$$

by integrating r derivatives over a cube of side $|h|$. The Hölder bound on $f^{(r)}$ therefore gives the difference criterion at order s , and the higher-order differences follow by applying additional differences. This proves the noninteger identification. Density transfers the little-Hölder convention. For $f \in C^r$, $r \geq 1$, (A.10) is $o(|h|^r)$, proving (2.6).

B Source provenance and the exact scope of reuse

The repository was inspected at commit

a21208b3ff14a07a4c8318dbf916d543acbef043.

The principal baseline is the article under

[Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Sharp_Cr_Spectral_Disks_Rvachev_Thue_Morse/article.tex](#)

Its Git blob is **45a9a8724aa42f2f6812d39bb486211438dcb09f**. The same source blob was observed during an independent inspection of a later branch head; all references in this package are pinned to the commit above. A file named **provenance.json** records the exact paths, blob identities, and the starting questions.

The baseline supplies the following starting material: the normalization $\mathcal{L} = \mathcal{P}/2$; the known three-mode eigenfunctions; the integer C^r result; the endpoint-jet algebra; and the explicitly stated questions about Hölder, Zygmund, and finite trigonometric masks. The present paper

reproves the finite Fourier identities and supplies the approximation quotient, the all-positive-order classification, the general boundary Jordan theorem, and the exact residual estimates.

The cited literature supplies context and standard analytic precedents. The proof in [appendix A](#) makes the required approximation facts explicit. No unreviewed repository claim about a different operator or an unrelated area is used. No fresh Lean or Rocq build was run, and no machine-checked status is attached to these new spectral theorems.

C Reproducibility and finite verification

The package is self-contained for compilation and finite verification. The LaTeX source includes its figure and bibliography, so no external graphics or bibliography database are required. A conventional TeX installation with the listed packages can compile it with `pdflatex`; three passes settle the table of contents and cross-references. The supplied `Makefile` performs this build.

From the verification directory, run

```
python3 verify.py -output results.json
```

The script uses only Python’s standard library, including `fractions.Fraction`. Its checks are explicit exceptions and remain enabled under optimized Python. No floating-point eigensolver, symbolic package, or network access is required.

Exact finite component	Successful checks
Three-dimensional sine core	21
Seven-dimensional higher-frequency core	22
Degree and affine-index identities	46,752
Defects, right inverse, and truncated critical series	332
Total	47,127

The degree grid contains 7,776 parameter tuples with $2 \leq b \leq 7$, $0 \leq d \leq 15$, $K \leq N \leq K + 8$, and $0 \leq n \leq 8$, plus 96 checks of the core degree. Some formal degrees in this grid cannot occur as exact degrees of normalized masks; they are used only to verify the integer identity and are not asserted to define admissible weights. Polynomial defects are checked through index 32, and the finite iterate identity is checked for every $0 \leq n \leq M \leq 16$.

The characteristic polynomials are computed by exact Faddeev–LeVerrier arithmetic. The proposed minimal polynomials are checked by matrix annihilation, necessity of each irreducible factor, and squarefreeness. Separate exact ranks verify the dimensions of the 1 and $-1/2$ eigenspaces. All matrix columns are compared with a separate sparse Fourier-polynomial implementation.

These are supplementary finite certificates. The norm estimates, series convergence, entire essential disks, and generalized-eigenvector exhaustion are established by the mathematical proofs, not by an extrapolation from the tested parameter grid.

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